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Sigma 9.0 (C++ DSA)

10 %

11. Patterns

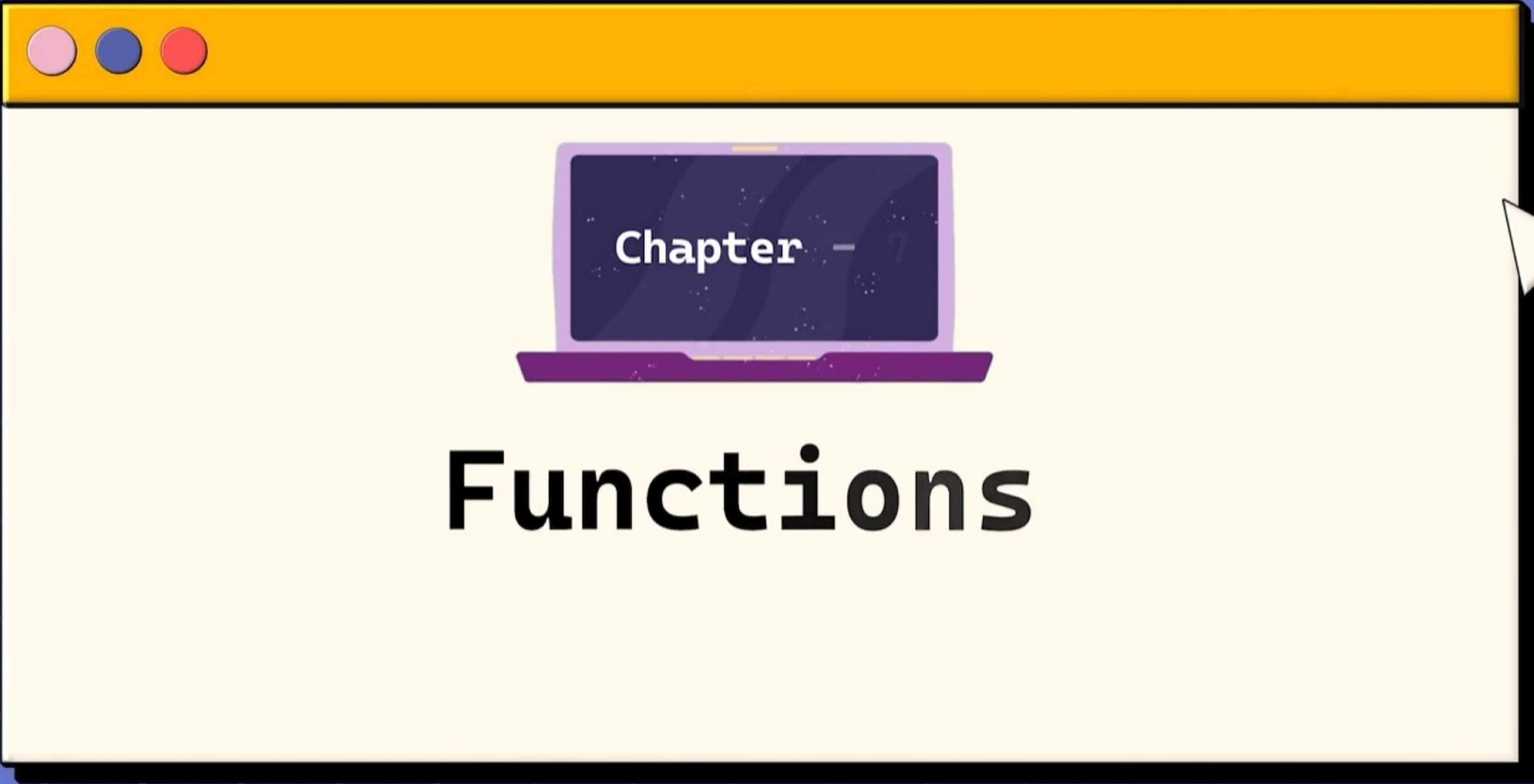
12. Functions

- What are Functions? video 09:03
- Forward Declaration video 03:31
- Syntax with Parameters video 07:29
- Practice Qs video 02:56
- Practice Qs (Factorial) video 05:14
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- Scope (Local & Global) video 05:31
- Practice Qs (Prime Number) video 05:48
- Practice Qs (Binomial Coefficient) video 05:09
- Function Overloading

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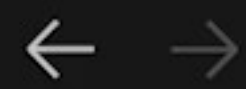
bood myFunction

Functions

Block of code which runs when it is called.

```
returnType fName () {  
    //do some work  
    return someValue; //optional  
}  
  
fName(); //function call
```





Cpp Codes



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code.cpp

code.cpp > sayHello()

```
1  #include <std::__1::ostream &std::__1::operator<<std::__1::char_traits<char>>
2  using namespace std;
3
4  void sayHello() {
5      cout << "Hello :) \n";
6  }
7
8  int main() {
9      sayHello(); //function call
10     return 0;
11 }
12
13
```

+19 overloads

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PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
○ apnacollege@Amans-MacBook-Pro Cpp Codes %
```



Example:

```
void sayHello() {  
    cout << "Hello :) \n";  
}  
  
int main() {  
    sayHello();  
    return 0;  
}
```

Handwritten annotations: A circled '2' is next to the `sayHello()` function definition. A circled '1' and '3' are next to the `sayHello();` and `return 0;` lines in the `main()` function, respectively. An arrow points from the `sayHello()` call in `main()` to the `sayHello()` function definition. Another arrow points from the `sayHello()` function definition to the `sayHello();` line in `main()`.



code.cpp ×

code.cpp > assistant()

```
1  #include <iostream>
2  using namespace std;
3
4  void sayHello() {
5      cout << "Hello :)\n";
6  }
7
8  void assistant() {
9      sayHello();
10     cout << "work done \n";
11 }
12
13 int main() {
14     assistant();
15     return 0;
16 }
17
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
Hello :)
work done
○ apnacollege@Amans-MacBook-Pro Cpp Codes %
```



Forward Declaration

Declaration: the function's name, return type, and parameters (if any)

Definition: the body of the function

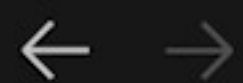
```
void sayHello() {  
    cout << "Hello :) \n";  
}
```

Declaration



Definition





Cpp Codes



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code.cpp x

code.cpp > ...

```
1  #include <iostream>
2  using namespace std;
3
4  void sayHello(); //function declare
5
6  int main() {
7      sayHello();
8      return 0;
9  }
10
11 void sayHello() {
12     cout << "Hello :)\n";
13 }
14
15
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
code.cpp:6:5: error: use of undeclared identifier 'sayHello'
    sayHello();
    ^
```

1 error generated.

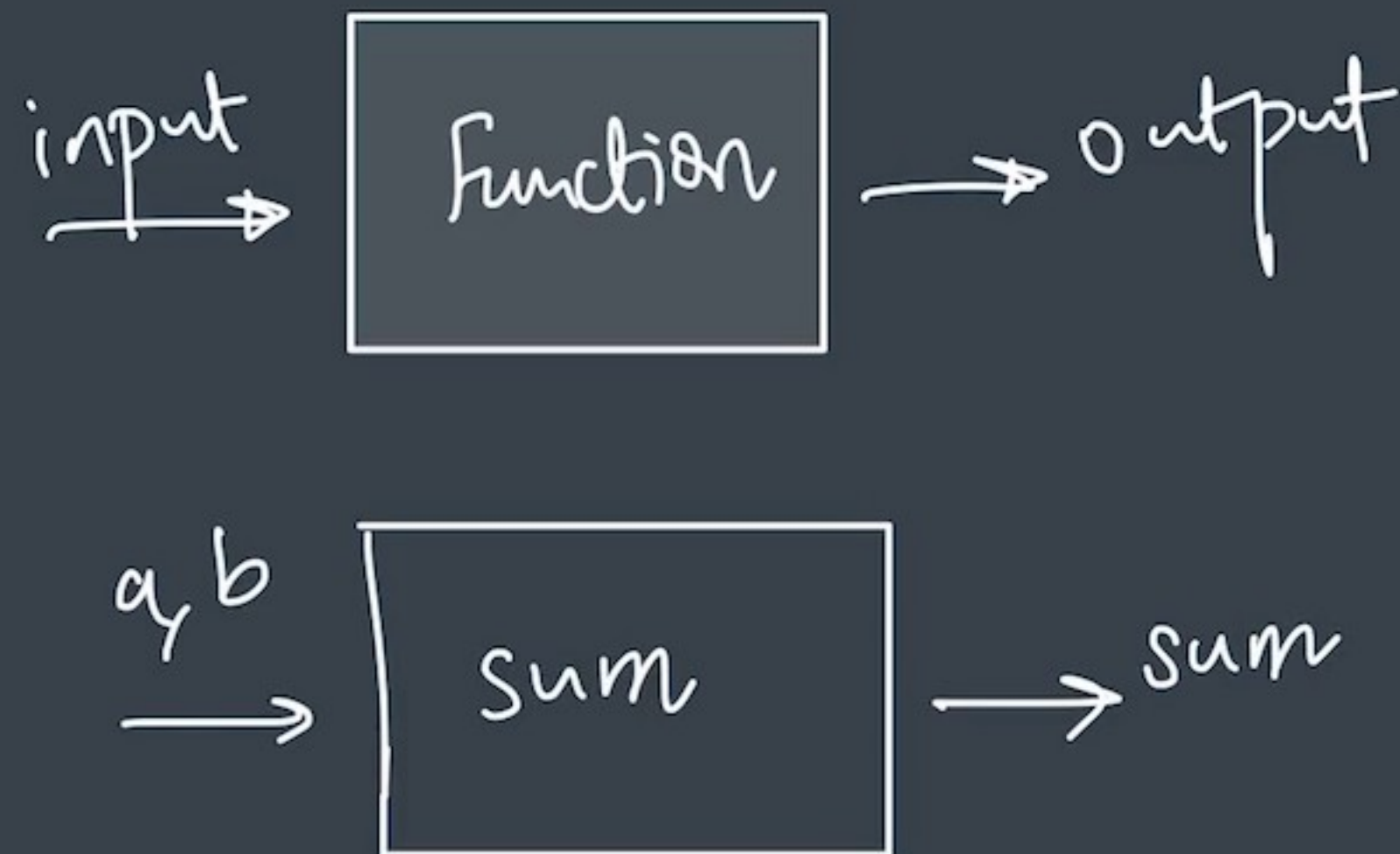
```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
Hello :)
```

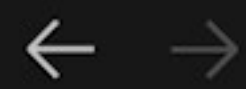
```
apnacollege@Amans-MacBook-Pro Cpp Codes % arman Khan 082020@gmail.com
```



Syntax with Parameters

```
returnType fname ( type param1, type param2 ..) {  
    //do some work  
    return someValue;  
}
```





Cpp Codes



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code.cpp

code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int sum(int a, int b) { // a, b are parameters
5      int sum = a + b;
6      return sum;
7  }
8
9  int main() {
10     int s = sum(2, 4);
11     cout << "sum = " << s << endl;
12     return 0;
13 }
14
15
```

PORTS

PROBLEMS

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code.cpp x

code.cpp > main()

```

8
9  int diff(int a, int b) { // a, b are parameters
10     int diff = a - b;
11     return diff;
12 }
13
14 int main() {
15     int s = sum(2, 4); //2,4 are arguments
16     cout << "sum = " << s << endl;
17     int d = diff(4, 2); //2,4 are arguments
18     cout << "diff = " << d << endl;
19     return 0;
20 }
21
22

```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```

● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
sum = 6
diff = 2
○ apnacollege@Amans-MacBook-Pro Cpp Codes %

```



code.cpp ×

code.cpp > sum(int, int)

```
1  #include <iostream>
2  using namespace std;
3
4  int sum(int a, int b=1) { // a, b are parameters
5      int sum = a + b;
6      return sum;
7  }
8
9  int diff(int a, int b) { // a, b are parameters
10     int diff = a - b;
11     return diff;
12 }
13
14 int main() {
15     int s = sum(2);
16     cout << "sum = " << s << endl;
17     return 0;
18 }
```

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Filter (e.g. text, **/*.ts, !**/node_

No problems have been detected in the workspace.



code.cpp ×

code.cpp > sum(int, int)

```
1  #include <iostream>
2  using namespace std;
3
4  int sum(int a=1, int b) { // a, b are parameters
5      int sum = a + b;
6      return sum;
7  }
8
9  int diff(int a, int b) { // a, b are parameters
10     int diff = a - b;
11     return diff;
12 }
13
14 int main() {
15     int s = sum(2);
16     cout << "sum = " << s << endl;
17     return 0;
18 }
```

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code.cpp:4:5: note: candidate function not viable: requires 2 arguments, but 1 was provided
int sum(int a=1, int b) { // a, b are parameters
^

2 errors generated.

apnacollege@Amans-MacBook-Pro Cpp Codes %



Practice Qs

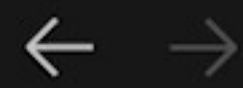
Qs. WAF to find the product of 2 numbers - a & b.

```
int prod(int a, int b) {  
    a * b  
}
```

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Qs. WAF to print if a number is odd or even.





Cpp Codes



code.cpp x

code.cpp > main()

```
8 //even -> true; odd -> false
9 bool isEven(int n) {
10     if(n % 2 == 0) {
11         return true;
12     } else {
13         return false;
14     }
15 }
16
17 int main() {
18     cout << isEven(20) << endl;
19     return 0;
20 }
21
22
```

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```
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
1
○ apnacollege@Amans-MacBook-Pro Cpp Codes %
```



Practice Qs

Qs. WAF to print the **factorial** of a number, n.

$$n! \Rightarrow \underbrace{1 \times 2 \times 3 \times 4 \dots \times n}$$

$$0! \Rightarrow 1$$

$$1! \Rightarrow 1$$

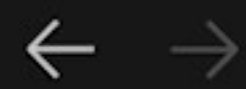
$$2! \Rightarrow 1 \times 2 = 2$$

$$3! \Rightarrow 1 \times 2 \times 3 \Rightarrow 6$$

$$4! \Rightarrow 1 \times 2 \times 3 \times 4 \Rightarrow 24$$

$$5! \Rightarrow 120$$





code.cpp

code.cpp > factorial(int)

```
13     return false;
14 }
15 }
16
17 int factorial(int n) {
18     int fact = 1;
19     for(int i=1; i<=n; i++){
20         fact = fact * i;
21     }
22     return fact;
23 }
24
25
26
27
28 int main() {
```

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PROBLEMS

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code.cpp

code.cpp > factorial(int)

```

16
17 int factorial(int n) {
18     int fact = 1;
19     for(int i=1; i<=n; i++) {
20         fact = fact * i;
21     }
22     cout << "factorial(" << n << ") = " << fact << endl;
23     return fact;
24 }
25

```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

zsh +

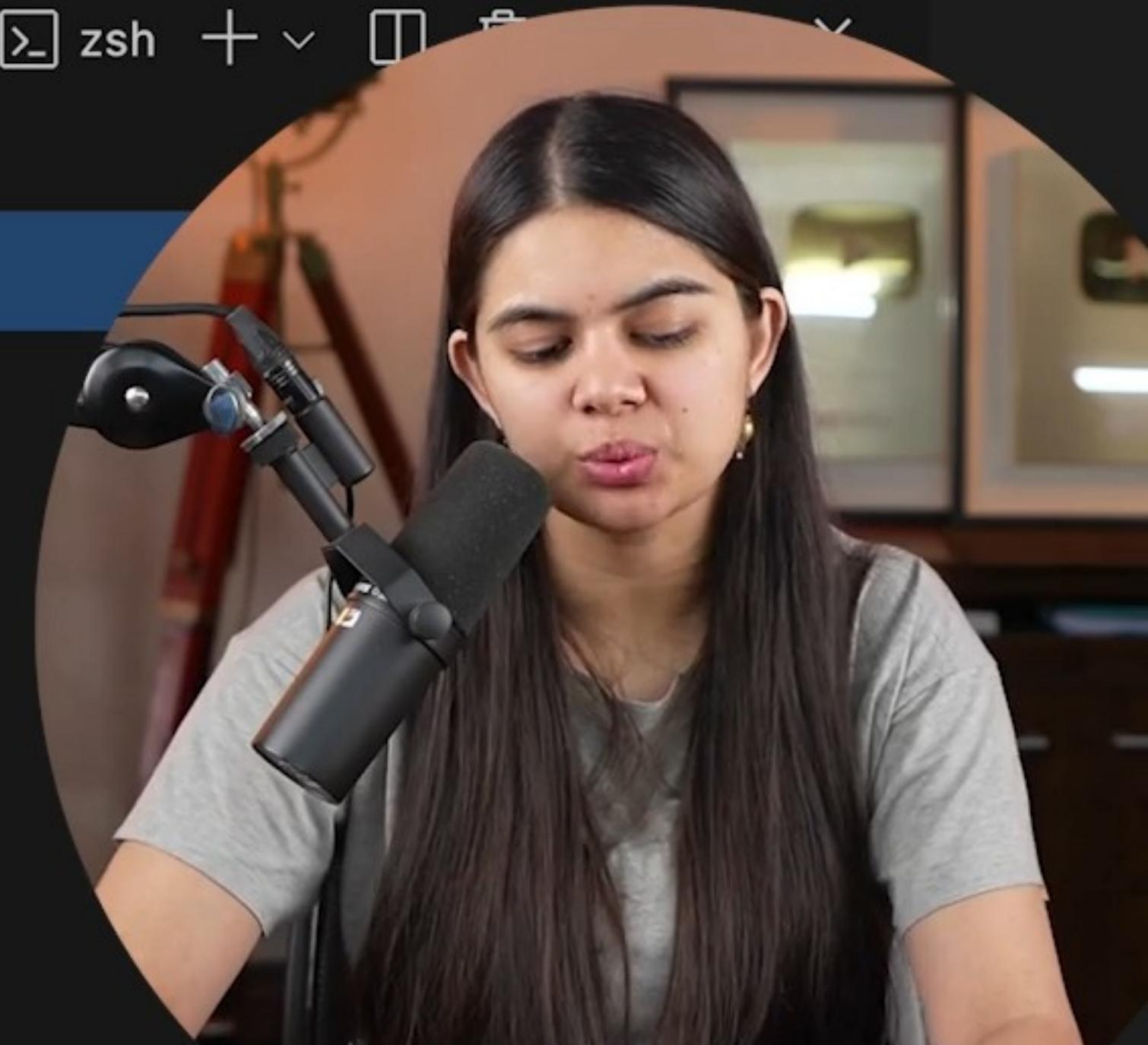
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

```

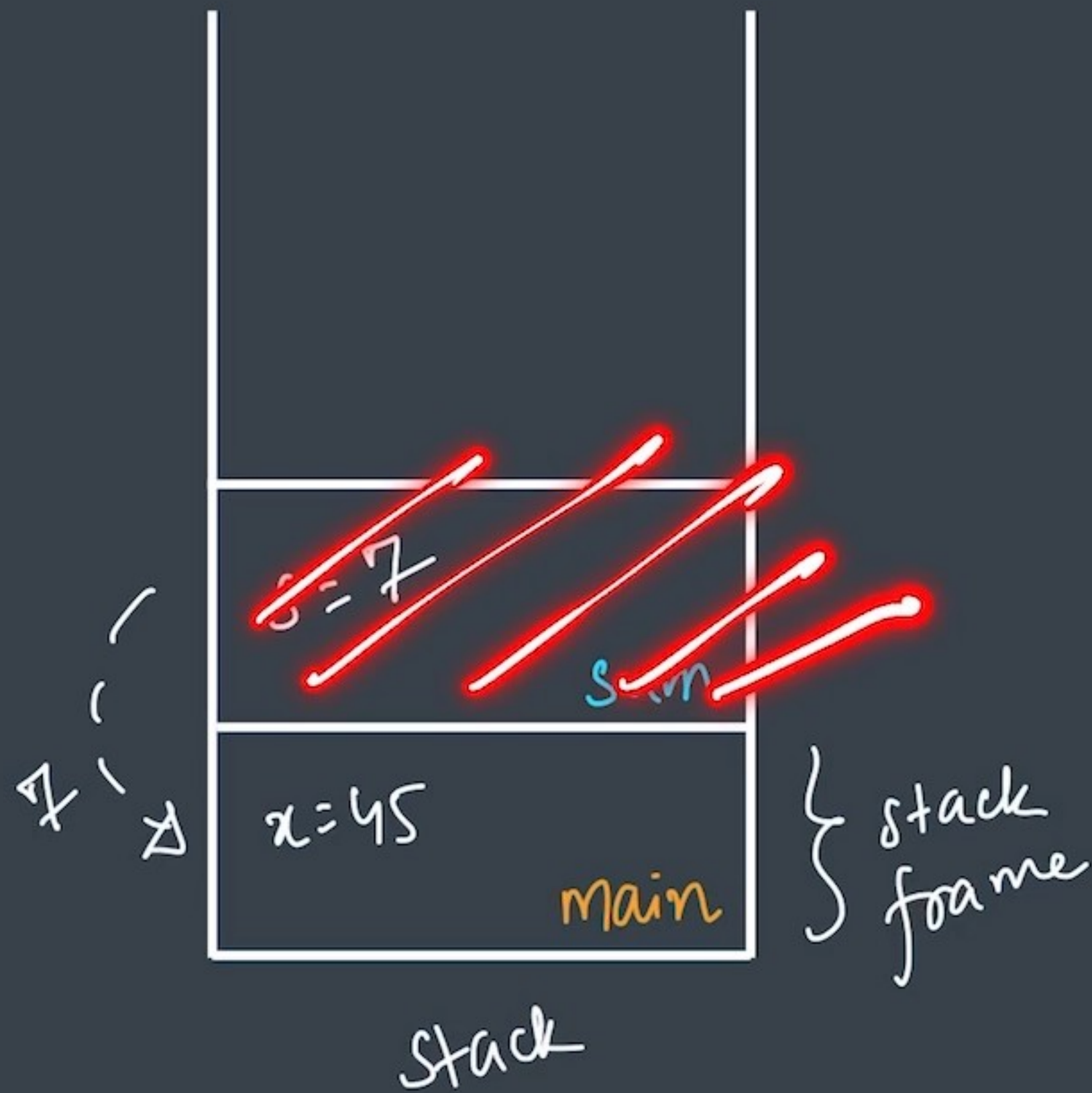
factorial(0) = 1
factorial(1) = 1
factorial(2) = 2
factorial(3) = 6
factorial(4) = 24
factorial(5) = 120
factorial(6) = 720

```

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What happens in **Memory**?



```
int sum(int a, int b) {  
    int s = a + b;  
    return s;  
}  
  
int main() {  
    int x = 45;  
    cout << sum(2, 5);  
    return 0;  
}
```

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What happens in **Memory?**



* current function
is at the top of
the stack.

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```
int sum(int a, int b) {  
    int s = a + b;  
    return s;  
}  
  
int main() {  
    int x = 45;  
    cout << sum(2, 5);  
    return 0;  
}
```



Scope

area where
a var can be accessed
or used.

Local vs Global scope of variables

- function →
- loops (i, j)
- if-else
- { }

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code.cpp 1 x

code.cpp > sum(int, int)

```
1  #include <iostream>
2  using namespace std;
3
4  void sum(int a, int b) {
5      int s = a + b;
6      cout << s;
7  }
8
9  int main() {
10     sum(5, 4);
11     cout << s;
12     return 0;
13 }
14
15
16
```

identifier "s" is undefined C/C++(20)

[View Problem \(⌘F8\)](#) [Quick Fix... \(⌘.\)](#)

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1

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code.cpp 1

code.cpp > sum(int, int)

```
1  #include <iostream>
2  using namespace std;
3
4  void sum(int a, int b) {
5      if(a >=
6          int
7      }
8      cout << x;
9      int s = a + b;
10     cout << s;
11 }
12
13 int main() {
14     sum(5, 4);
15     int s = 10;
16     cout << s;
17     return 0;
```

identifier "x" is undefined C/C++(20)

View Problem (\F8) Quick Fix... (\%.)

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code.cpp 1

code.cpp > sum(int, int)

```
1  #include <iostream>
2  using namespace std;
3
4  void sum(int a, int b) {
5      {
6          int
7      }
8      cout << x;
9      int s = a + b;
10     cout << s;
11 }
12
13 int main() {
14     sum(5, 4);
15     int s = 10;
16     cout << s;
17     return 0;
```

identifier "x" is undefined C/C++(20)

View Problem (\F8) Quick Fix... (\%.)

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PROBLEMS

1

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code.cpp ×

code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int num = 25;
5
6  void sum(int a, int b) {
7      cout << num;
8      int s = a + b;
9      cout << s;
10 }
11
12 int main() {
13     sum(5, 4);
14     int s = 10;
15     cout << num;
16     return 0;
17 }
```

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PROBLEMS

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```
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
25925%
○ apnacollege@Amans-MacBook-Pro Cpp Codes %
```



code.cpp ×

code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int num = 25;
5
6  void sum(int a, int b) {
7      cout << num << endl;
8      int s = a + b;
9      cout << s << endl;
10 }
11
12 int main() {
13     sum(5, 4);
14     int s = 10;
15     cout << num << endl;
16     return 0;
17 }
```

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PROBLEMS

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25925%

● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

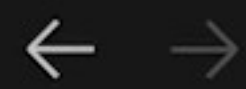
25

9

25

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Cpp Codes



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code.cpp x

code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  bool isPrime(int n) {
5      if(n == 1) {
6          return false;
7      }
8
9      for(int i=2; i<=n-1; i++) {
10         if(n % i == 0) { //non-prime
11             return false;
12         }
13     }
14
15     return true;
16 }
17
```

PORTS

PROBLEMS

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Practice Qs

Qs. WAF to print if a number is **prime** or not.

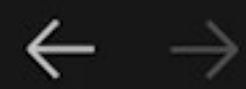
$$\begin{aligned} \underline{i \leq \sqrt{n}} &\downarrow \\ i^2 \leq n \\ i \times i \leq n &\uparrow \end{aligned}$$

1 → non prime

```
for (int i = 2; i <= n-1; i++) {
    (n % i) → non-prime (false)
}
```

ans → prime (true)





Cpp Codes



code.cpp

code.cpp > isPrime2(int)

```
15     return true;
16 }
17
18 bool isPrime2(int n) {
19     if(n == 1) {
20         return false;
21     }
22
23     for(int i=1; i*i<=n; i++) {
24         if(n % i == 0) {
25             return false;
26         }
27     }
28
29     return true;
30 }
31
```

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```
0
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
1
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
1
○ apnacollege@Amans-MacBook-Pro Cpp Codes %
```



code.cpp ×

code.cpp > main()

```

19     if(n == 1) {
20         return false;
21     }
22
23     for(int i=2; i*i<=n; i++) {
24         if(n % i == 0) {
25             return false;
26         }
27     }
28
29     return true;
30 }
31
32 int main() {
33     cout << isPrime2(2) << endl;
34     return 0;
35 }

```

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```

● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
1
○ apnacollege@Amans-MacBook-Pro Cpp Codes %

```

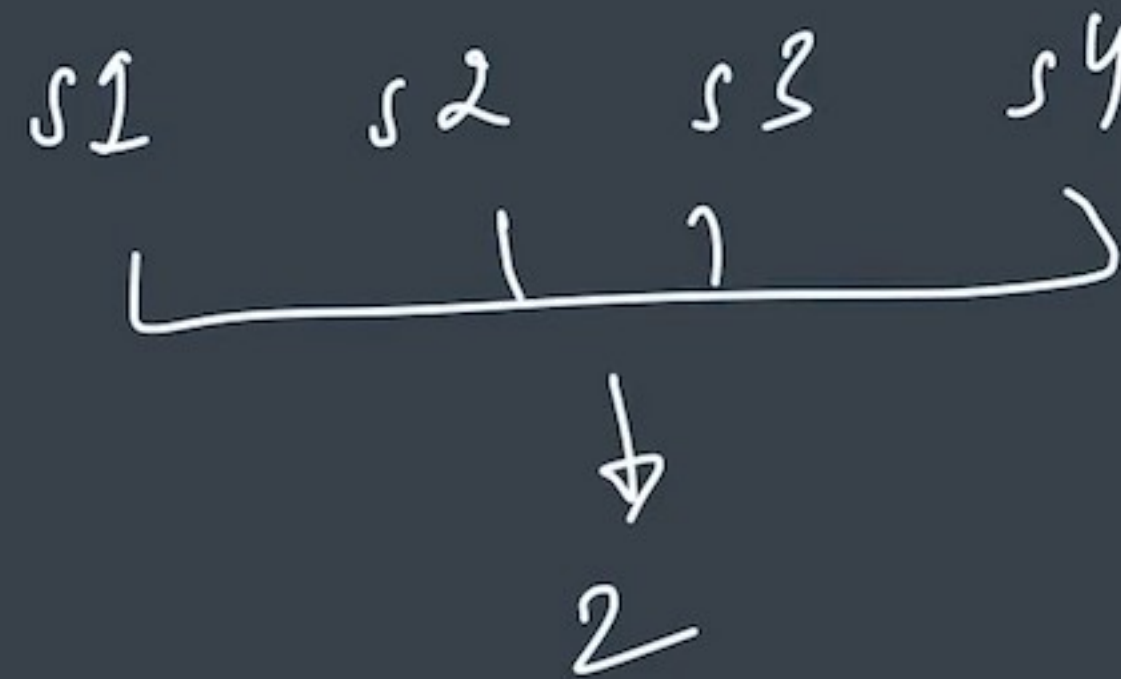


Practice Qs

Qs. WAF to find the **binomial coefficient** for given n & r.

$${}^nC_r = \frac{n!}{r! (n-r)!}$$

4



s1, s2

s1, s3

s1, s4

s2, s3

s2, s4

s3, s4

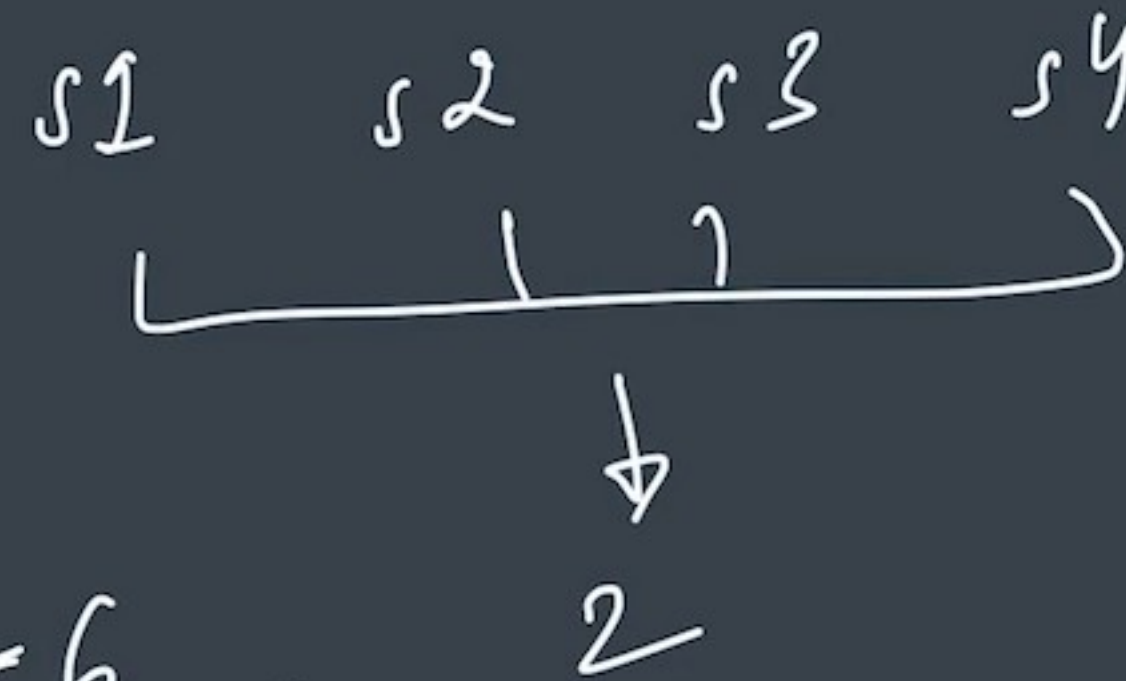
6 choices



Practice Qs

Qs. WAF to find the **binomial coefficient** for given n & r.

$${}^n C_r = \frac{n!}{r! (n-r)!}$$

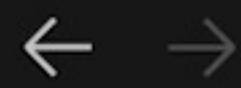


$${}^4 C_2 = \frac{4!}{2! 2!} = \frac{24}{2 \times 2} = 6$$

- s1, s2
- s1, s3
- s1, s4
- s2, s3
- s2, s4
- s3, s4

6 choices





Cpp Codes



code.cpp ×

code.cpp > main()

```
12
13  int binCoeff(int n, int r) {
14      int val1 = factorial(n);
15      int val2 = factorial(r);
16      int val3 = factorial(n-r);
17
18      int result = val1 / (val2 * val3);
19      return result;
20  }
21
22  int main() {
23      cout << binCoeff(4, 2) << endl;
24      return 0;
25  }
26
27
```

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PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
6
○ apnacollege@Amans-MacBook-Pro Cpp Codes % █
I
```





Function Overloading

Multiple functions with the same name but different parameters

sum ()

sum ()



code.cpp

code.cpp > sum2(double, double)

```
1  #include <iostream>
2  using namespace std;
3
4  int sum(int a, int b) {
5      cout << (a+b) << endl;
6      return a + b;
7  }
8
9  double sum2(double a, double b) {
10     return a + b;
11 }
12
13 int main() {
14     sum(2, 3);
15     sum(12, 13);
16     sum(1.5, 2.5); //4
17     return 0;
18 }
19
20
```



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Function Overloading

Multiple functions with the same name but different parameters

`sum (int a, int b)`

`sum (int a, int b, int c)`

type

no. of parameters



Practice Qs

Qs. Print all Primes in a Range from 2 to n.

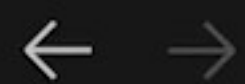
```
allPrimes (int n) {
```

```
  for (i=2 to n)
```

```
    isPrime(i) }
```

}





code.cpp ×

code.cpp > ...

```
10  
17  
18 void allPrimes(int n) {  
19     for(int i=2; i<=n; i++) {  
20         if(isPrime(i)) { //true  
21             cout << i << " ";  
22         }  
23     }  
24  
25     cout << endl;
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

zsh + ▾ ▢ ▢ ... ^ ×

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out  
2 3 5 7 11 13  
apnacollege@Amans-MacBook-Pro Cpp Codes %
```

